

TITLE PAGE

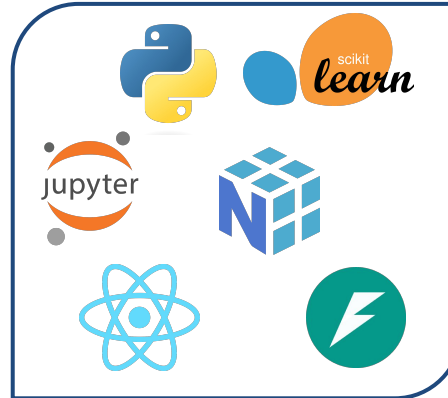
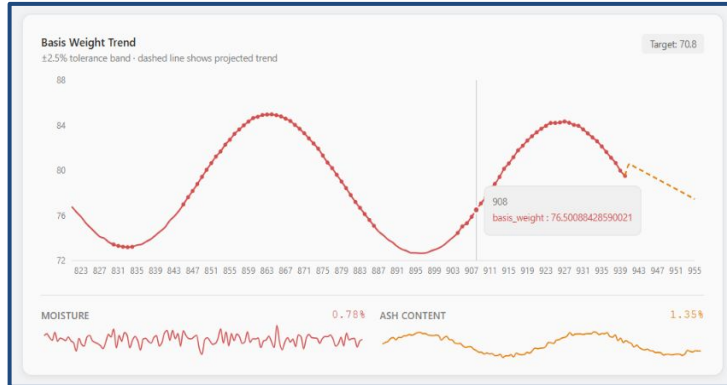
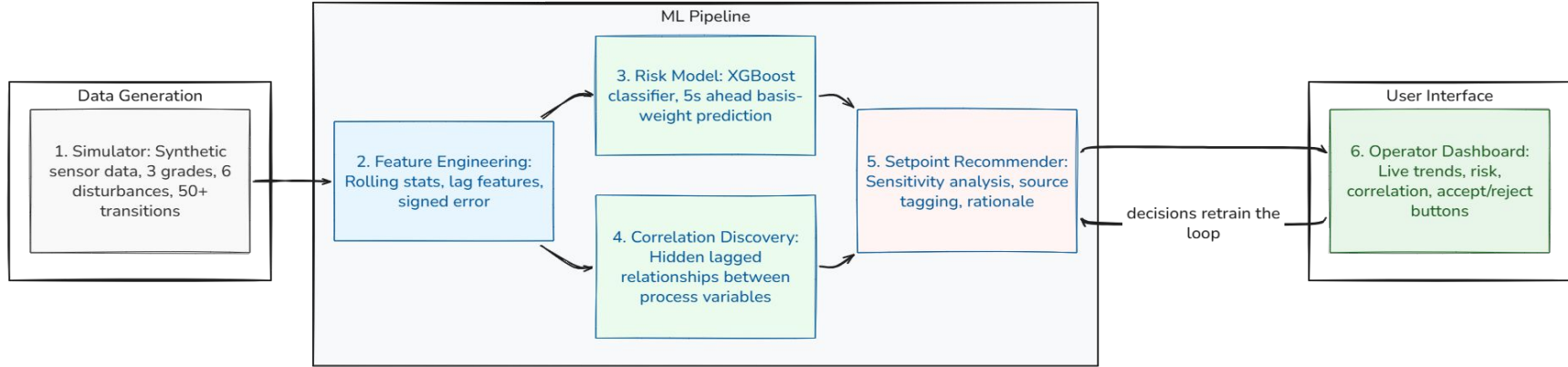
- **Problem Statement ID – 5**
- **Problem Statement Title- Grade Change Intelligence in Paper Making Process**
- **Theme- Paper Manufacturing, Control Systems**
- **PS Category- Software**
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IDEA TITLE

Grade Change Intelligence in Paper Making Process

- Layer additional to Honeywell's existing **QCS MD Control**, to assist with operator recommendations and MD Grade changes. Predicts basis-weight risk **before** crossing **$\pm 2.5\%$ spec band**.
- **Off-spec risk model** flags trouble 5 seconds ahead with **98.8% accuracy** on unseen transitions, **versus 86.5% for a naive "same as now" guess**, containing rationale and source tag.
- Physics-informed **digital twin synthetic data simulator** (mass balance + FOPDT + dynamic dead time) generates the training data, so results are traceable to real process behavior, not a black box.
- **Augments the control loop**, as a **low-risk add on** for mills running on QCS.
- **Closed feedback loop**: accept/reject decisions get logged and checked against the real outcome, affecting real parameter suggestions with **Source-of-inference tagging**.

TECHNICAL APPROACH



Tech Stack

- **Simulation:** NumPy, pandas
- **Modeling:** XGBoost, scikit-learn
- **Backend:** FastAPI, uvicorn
- **Frontend:** React + Vite, Recharts
- **Storage:** CSV and JSON files

FEASIBILITY AND VIABILITY

PROTOTYPE FEASIBILITY

Prototype with Valid Metrics having Physics and Mathematical constraints

Non-Invasive - No Hardware Changes

Standard Stack - XGBoost and FastAPI

Lightweight - No MPC Solver Required

- Due to **True Randomness** not being existent in Computer Science, it is difficult to simulate **Real Historian Data as QCS history, MIS reports, DCS historian trends** are not open sourced.
- This system when Integrated with **Honeywell Experion MX and Alpha MPC** can build **operator trust** and adoption of **automated suggestions**.

Strategies for overcoming these challenges:

- **Phased rollout:** shadow mode first (log only), then advisory (operator decides), only later supervised auto-apply
- **Stratified accuracy reporting** keeps the weak spots visible.

RISKS AND CHALLENGES

Data Noise - Raw Historian vs Simulated

Tuning - Recommendation Delta Sizing

Recipe Mismatch - Lower Accuracy 98.5%

Human Factor - Operator Trust and Adoption

MITIGATION STRATEGIES

Phased Rollout - Shadow to Advisory to Auto

Feedback Loop - Accept or Reject Retraining

Transparency - Stratified Accuracy Reports

Safety First - Human-in-the-Loop Design

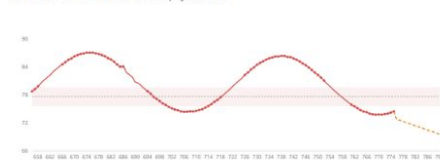
Grade Change Intelligence

Live basis-weight monitoring & AI setpoint recommendations

Basis Weight Trend

±2.5% tolerance band - dashed line shows projected trend

Target: 77.7



PLAY

1x

5x

20x

INDEX 700

Go

Off-Spec Risk

Risk prediction for next 5 seconds

100.0 %

Critical

GRADE

Grade B

TRANSITION

#0

STATE

Transitioning

NOW

Off-spec

Discovered Correlations

Variable relationships and inference sources

dryer_temp 30s → moisture $r=-0.96$ cross-correlation

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Variable relationships and inference sources

dryer_temp 30s → moisture $r=-0.96$ cross-correlation

steam_pressure 10s → filler_flow $r=0.95$ cross-correlation

machine_speed 10s → steam $r=-0.95$ cross-correlation

dryer_temp 10s → filler_flow $r=0.95$ cross-correlation

machine_speed 10s → dryer_ $r=-0.95$ cross-correlation

dryer_temp 10s → moisture $r=-0.95$ cross-correlation

FEATURE IMPORTANCE

machine_speed_1s

basis_weight_rm

basis_weight_rm

basis_weight_rm

machine_speed_1s

stock_flow_lag60

DEVIATION BY SCENARIO

Recipe Mismatch 25.28% avg - 303.3% max

Disturbed Trans. 11.92% avg - 124.91% max

Slow Actuator 11.3% avg - 105.3% max

Steam Lag 11.25% avg - 114.7% max

Normal 11.16% avg - 102.79% max

Aggressive Operator 9.44% avg - 110.84% max

Off-Spec Risk

Risk prediction for next 5 seconds

12.1 %

Normal

GRADE

Grade B

TRANSITION

#0

STATE

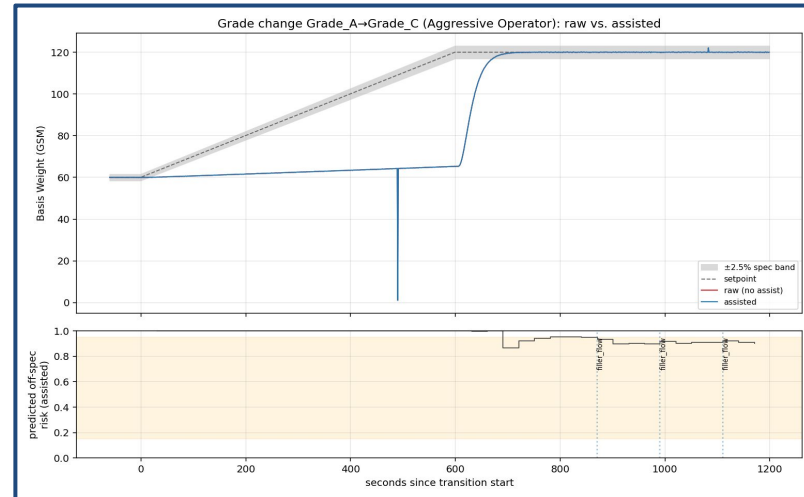
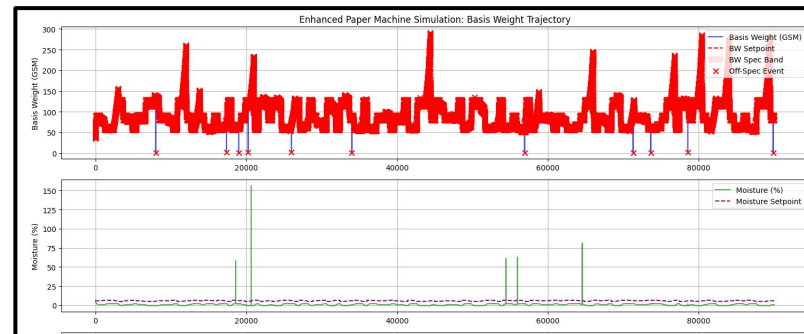
Transitioning

NOW

In-spec

ARTIFACTS

Code snippets Attached in Zip File



RESEARCH AND REFERENCES

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